

Scenario

You are a data engineer hired by a solid waste management company. It collects and recycles solid waste across major cities in the country of Brazil. They operate hundreds of trucks of different types to collect and transport solid waste. The company would like to create a data warehouse so that it can create reports like:

- Total waste collected per year per city
- Total waste collected per month per city
- Total waste collected per quarter per city
- Total waste collected per year per trucktype
- Total waste collected per trucktype per city
- Total waste collected per trucktype per station per city

You will use your data warehousing skills to design and implement a data warehouse for the company.

Exercise 1: Design a data warehouse

The solid waste management company has provided you the sample data they want to collect.

Trip number	Waste Type	Waste Collected in tons	Collection Zone	City	Date
1	Dry	45.23	South	Sao Paulo	23-Jan-20
2	Wet	43.12	Central	Rio de Janeiro	24-Jan-20
3	Electronic	40.19	South	Sao Paulo	23-Jan-20
4	Plastic	34.87	West	Rio de Janeiro	24-Jan-20
5	Wet	45.34	West	Rio de Janeiro	23-Jan-20

You will start your project by designing a Star Schema warehouse by identifying the columns for the various dimensions and fact tables in the schema.

Task 1: Design the dimension table MyDimDate

Write down the fields in the MyDimDate table in any text editor, one field per line. The company is looking at a granularity of day, which means they would like to have the ability to generate the report on a yearly, monthly, daily, and weekday basis.

dateid -> surrogate key (PK)
fulldate -> actual calendar date
day -> day of month, 1-31
month -> month number, 1-12
monthname -> 'January', 'February', ...
quarter -> 1-4
quartername -> 'Q1', 'Q2', 'Q3', 'Q4'
year -> e.g. 2019
weekday -> 1-7
weekdayname -> 'Monday', 'Tuesday', ...

Task 2: Design the dimension table MyDimWaste

Write down the fields in the MyDimWaste table in any text editor, one field per line.

wasteid -- surrogate key (PK)
trucktype -- e.g. 'Volvo', 'Scania'
wastecategory -- e.g. 'General', 'Recyclable', 'Organic'

Task 3: Design the dimension table MyDimZone

Write down the fields in the MyDimZone table in any text editor, one field per line.

zoneid -- surrogate key (PK)
stationid -- natural station identifier
city -- city the station operates in
region -- optional: state/region, if the company later provides it

Task 4: Design the fact table MyFactTrips

Write down the fields in the MyFactTrips table in any text editor, one field per line.

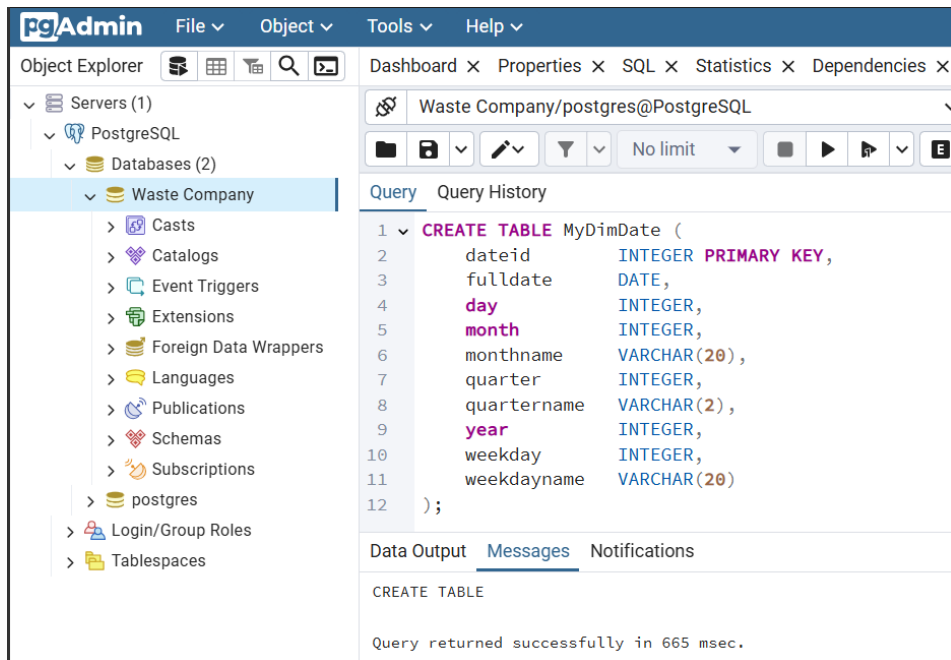
tripid -- surrogate key (PK)
dateid -- FK -> MyDimDate
zoneid -- FK -> MyDimZone
wasteid -- FK -> MyDimWaste
wastecollected -- numeric measure (e.g. tons)

Exercise 2 - Create schema for data warehouse on PostgreSQL

In this exercise, you will create the tables you have designed in the previous exercise. Open pgAdmin and create a database named **Project**, then create the following tables.

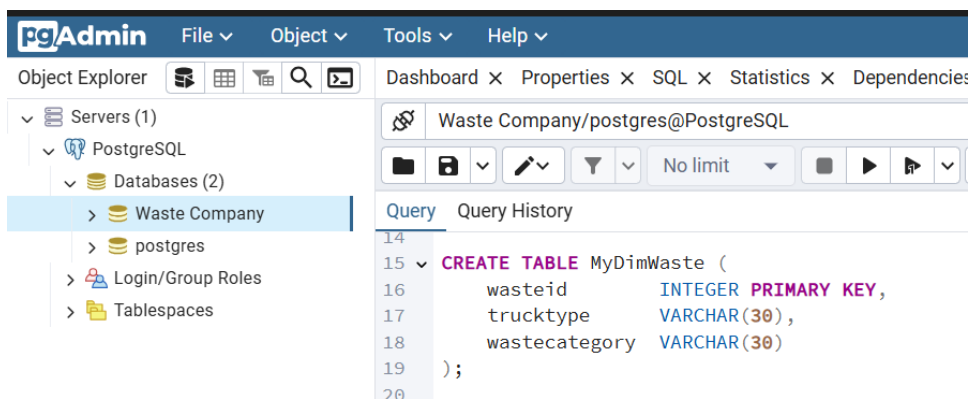
Task 5: Create the dimension table MyDimDate

Create the MyDimDate table.



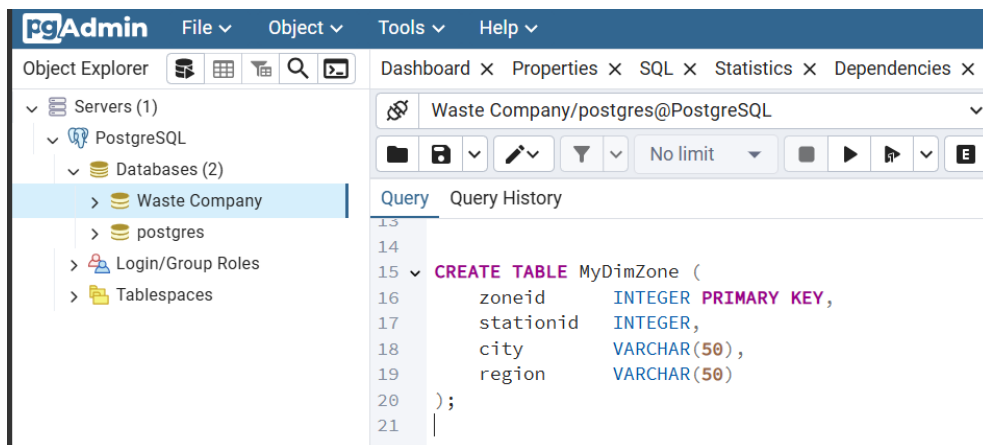
Task 6: Create the dimension table MyDimWaste

Create the MyDimWaste table.



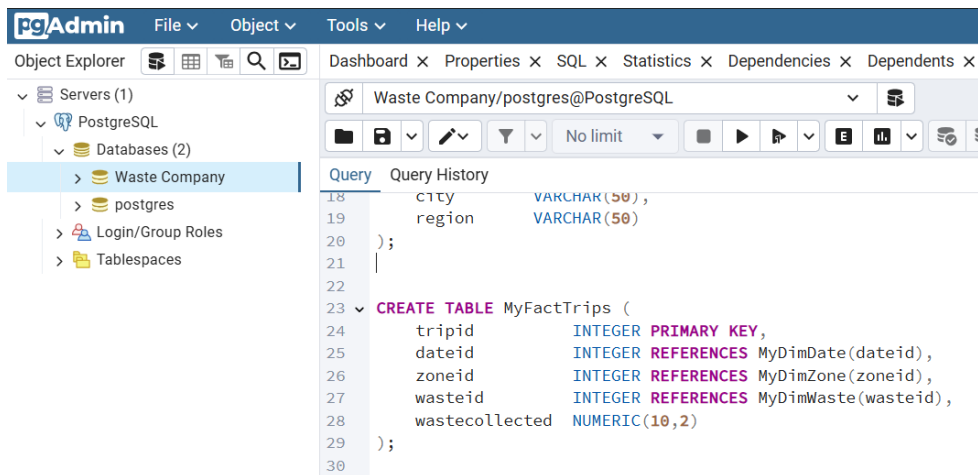
Task 7: Create the dimension table MyDimZone

Create the MyDimZone table.



Task 8: Create the fact table MyFactTrips

Create the MyFactTrips table.



Exercise 3: Load data into the data warehouse

In this exercise, you will load the data into the tables.

After the initial schema design, you were told that due to operational issues, data could not be collected in the format initially planned. This implies that the previous tables (MyDimDate, MyDimWaste, MyDimZone, MyFactTrips) in the *Project* database and their associated attributes are no longer applicable to the current design. The company has now provided data in CSV files with new tables DimTruck and DimStation as per the new design.

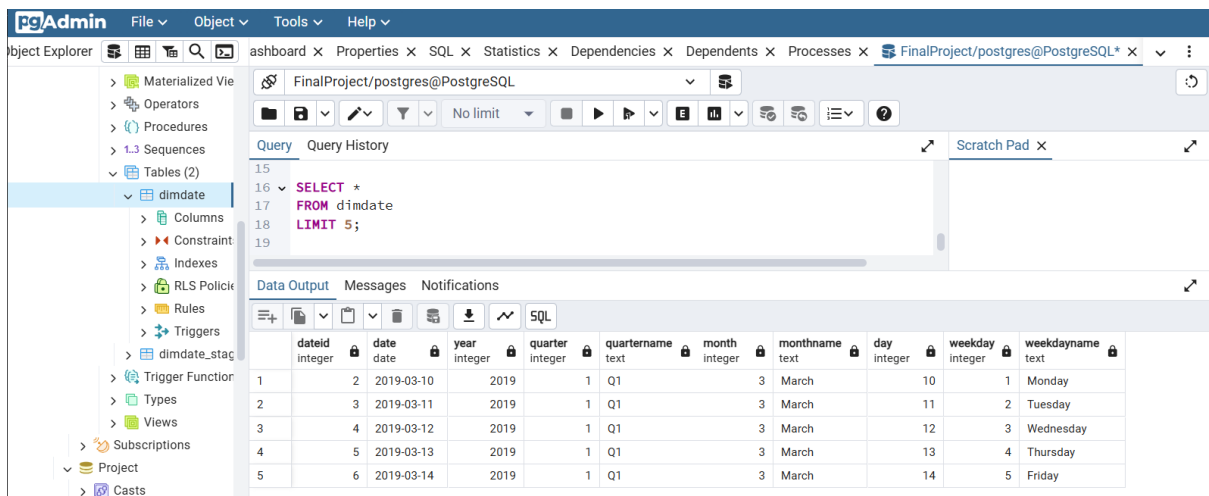
You will need to load the data provided by the company in CSV format. First, create a new database named **FinalProject**. Then, create the tables DimDate, DimTruck, DimStation, and FactTrips by defining the structure of the columns as per the CSV files. Next, load the data from the CSV files into the appropriate tables.

Note: Ensure that you upload the files to this path: /var/lib/pgadmin/

Task 9: Load data into the dimension table DimDate

Download the data from <https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0260EN-SkillsNetwork/labs/Final%20Assignment/DimDate.csv>

Load this data into DimDate table.



Task 10: Load data into the dimension table DimTruck

Download the data from <https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0260EN-SkillsNetwork/labs/Final%20Assignment/DimTruck.csv>

Load this data into DimTruck table.

Take a screenshot of the first 5 rows in the table DimTruck.

The screenshot shows the PgAdmin interface. On the left, the Object Explorer shows the database structure. The main pane displays the 'Query' tab with the following SQL query:

```
1 SELECT *
2 FROM dimtruck
3 LIMIT 5;
```

Below the query, the 'Data Output' tab shows the first 5 rows of the 'dimtruck' table:

	truckid [PK] integer	trucktype character varying (20)
1	115	Volvo
2	120	Scania
3	121	Volvo
4	122	Scania
5	125	Volvo

Task 11: Load data into the dimension table DimStation

Download the data from <https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0260EN-SkillsNetwork/labs/Final%20Assignment/DimStation.csv>

Load this data into DimStation table.

Take a screenshot of the first 5 rows in the table DimStation.

The screenshot shows the PgAdmin interface. On the left, the Object Explorer shows the database structure. The main pane displays the 'Query' tab with the following SQL query:

```
6
7 SELECT * FROM dimstation
8 LIMIT 5;
9
```

Below the query, the 'Data Output' tab shows the first 5 rows of the 'dimstation' table:

	stationid [PK] integer	city character varying (50)
1	19	Sao Paulo
2	21	Sao Paulo
3	31	Rio de Janeiro
4	32	Rio de Janeiro
5	40	Brasilia

Task 12: Load data into the fact table FactTrips

Download the data from <https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DB0260EN-SkillsNetwork/labs/Final%20Assignment/FactTrips.csv>

Load this data into FactTrips table.

Take a screenshot of the first 5 rows in the table FactTrips.

The screenshot shows the PgAdmin interface. On the left, the 'Object Explorer' pane shows the database structure, with 'facttrips' selected under 'Tables (8)'. The main pane displays the 'Query' editor with the following SQL query:

```
11  
12 SELECT *  
13 FROM FactTrips  
14 LIMIT 5;
```

Below the query editor, the 'Data Output' tab is active, showing the first 5 rows of the 'FactTrips' table. The table has the following columns: tripid (PK) integer, dateid integer, stationid integer, truckid integer, and wastecollected numeric (10,2).

	tripid [PK] integer	dateid integer	stationid integer	truckid integer	wastecollected numeric (10,2)
1	23475	1	71	133	33.36
2	23476	1	46	162	34.88
3	23477	1	40	134	34.69
4	23478	1	43	148	30.01
5	23479	1	46	169	37.47

Exercise 4 - Write aggregation queries and create materialized views

In this exercise, you will query the data you have loaded in the previous exercise.

Task 13: Create a grouping sets query

Create a grouping sets query using the columns stationid, trucktype, total waste collected.

Take a screenshot of the SQL and the output rows.

pgAdmin File Object Tools Help

Object Explorer

- Servers (1)
 - PostgreSQL
 - Databases (3)
 - FinalProject
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables

FinalProject/postgres@PostgreSQL

No limit

Data Output Messages Notifications

	stationid integer	trucktype character varying (20)	totalwastecollected numeric
1	19	Scania	248915.75
2	19	Volvo	83901.03
3	19	[null]	332816.78
4	21	Scania	248165.20
5	21	Volvo	84976.81
6	21	[null]	333142.01
7	31	Scania	83461.06
8	31	Volvo	83600.62
9	31	[null]	167061.68
10	32	Scania	125627.46
11	32	Volvo	42137.38
12	32	[null]	167764.84
13	40	Scania	82155.15

Task 14: Create a rollup query

Create a rollup query using the columns year, city, stationid, and total waste collected.

Take a screenshot of the SQL and the output rows.

pgAdmin File Object Tools Help

Object Explorer

- Servers (1)
 - PostgreSQL
 - Databases (3)
 - FinalProject**
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables

FinalProject/postgres@PostgreSQL

Query Query History

9 (ft.StationID),

Data Output Messages Notifications

	year integer	city character varying (50)	stationid integer	totalwastecollected numeric
1	2019	Brasilia	40	138850.54
2	2019	Brasilia	43	141503.54
3	2019	Brasilia	46	144418.78
4	2019	Brasilia	71	141856.38
5	2019	Brasilia	77	141627.10
6	2019	Brasilia	97	140485.97
7	2019	Brasilia	[null]	848742.31
8	2019	Rio de Janeiro	31	143821.05
9	2019	Rio de Janeiro	32	142766.60
10	2019	Rio de Janeiro	44	141766.33
11	2019	Rio de Janeiro	57	141405.86

Task 15: Create a cube query

Create a cube query using the columns year, city, stationid, and average waste collected.

Take a screenshot of the SQL and the output rows.

pgAdmin File Object Tools Help

Object Explorer

- FTS Parsers
- FTS Templates
- Foreign Tables
- Functions
- Materialized Views (1)
 - max_waste_stats
 - Columns
 - Indexes
- Operators
- Procedures
- 1..3 Sequences
- Tables (8)
 - dimdate
 - dimdate_stage
 - dimstation
 - dimstation_stage
 - dimtruck
 - dimtruck_stage
 - facttrips
 - facttrips_stage
- Trigger Functions

FinalProject/postgres@PostgreSQL

Query Query History

Data Output Messages Notifications

	city character varying (50)	stationid integer	trucktype character varying (20)	maxwastecollected numeric
1	Brasilia	40	Scania	45.00
2	Brasilia	40	Volvo	45.00
3	Brasilia	43	Scania	45.00
4	Brasilia	43	Volvo	44.99
5	Brasilia	46	Scania	45.00
6	Brasilia	46	Volvo	45.00
7	Brasilia	71	Scania	45.00
8	Brasilia	71	Volvo	45.00
9	Brasilia	77	Scania	45.00
10	Brasilia	77	Volvo	44.98
11	Brasilia	97	Scania	45.00